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Advanced Control System Design - Tutorial 3 – Solution

Find the state-space representations of the following systems using multiple methods:

Note that some steps have been skipped. The complete steps are available in the lecture's slides.

1. $\dfrac{Y(s)}{U(s)} = \dfrac{5s^2+7s+9}{s^3+8s^2+6s+2}$

***n*th order:**

$$\beta_0 = b_0 = 0$$

$$\beta_1 = b_1 - a_1\beta_0 = 5 - 8\times 0 = 5$$

$$\beta_2 = b_2 - a_1\beta_1 - a_2\beta_0 = 7 - 8\times 5 - 6\times 0 = -33$$

$$\beta_3 = b_3 - a_1\beta_2 - a_2\beta_1 - a_3\beta_0 = 9 - 8\times(-33) - 6\times 5 - 2\times 0 = 243$$

$$\dot{x}_1 = x_2 + \beta_1 u = x_2 + 5u$$

$$\dot{x}_2 = x_3 + \beta_2 u = x_3 - 33u$$

$$\dot{x}_3 = -a_3x_1 - a_2x_2 - a_1x_3 + \beta_3 u = -2x_1 - 6x_2 - 8x_3 + 243u$$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -2 & -6 & -8\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}5\\ -33\\ 243\end{bmatrix}u$$

$$y = \begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \beta_0 u = \begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

CCF:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -a_3 & -a_2 & -a_1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 1\end{bmatrix}u = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -2 & -6 & -8\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 1\end{bmatrix}u$$

$$y=\begin{bmatrix} b_3-a_3b_0 & b_2-a_2b_0 & b_1-a_1b_0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}+b_0u=\begin{bmatrix} 9 & 7 & 5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

OCF:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix}=\begin{bmatrix} 0 & 0 & -a_3 \\ 1 & 0 & -a_2 \\ 0 & 1 & -a_1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}+\begin{bmatrix} b_3-a_3b_0 \\ b_2-a_2b_0 \\ b_1-a_1b_0 \end{bmatrix}u=\begin{bmatrix} 0 & 0 & -2 \\ 1 & 0 & -6 \\ 0 & 1 & -8 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}+\begin{bmatrix} 9 \\ 7 \\ 5 \end{bmatrix}u$$

$$y=\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}+b_0u=\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

DCF:

The system has complex eigenvalues of -0.397±*j*0.346 and -7.206. Thus, this case is not examinable (not required). However, you may take a look at this solution for general knowledge:

The diagonal representation of the system can be given as follows:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix}=\begin{bmatrix} -p_1 & 0 & 0 \\ 0 & -p_2 & 0 \\ 0 & 0 & -p_3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}+\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}u$$

$$y=\begin{bmatrix} c_1 & c_2 & c_n \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}+b_0u$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix}=\begin{bmatrix} -0.397+j0.346 & 0 & 0 \\ 0 & -0.397-j0.346 & 0 \\ 0 & 0 & -7.206 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}+\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}u$$

$$y=\begin{bmatrix} 0.153-j1.367 & 0.153+j1.367 & 4.694 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

However, as was discussed before, such a complex representation is not suitable in several cases. Thus, it is required to find the real diagonal form. There are several approaches to

achieve this. One of them is done by transforming the **A** matrix of the complex diagonal representation ($\mathbf{A}_{CD}$) to a real representation ($\mathbf{A}_{RD}$), which can be represented as follows:

$$\mathbf{A}_{RD} = \begin{bmatrix} \sigma_1 & \omega_1 & 0 \\ -\omega_1 & \sigma_1 & 0 \\ 0 & 0 & p_3 \end{bmatrix}$$

Define the matrix **R**:

$$\mathbf{R} = \begin{bmatrix} 1 & 1 & 0 \\ j & -j & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Thus:

$$\mathbf{A}_{RD} = \mathbf{R}\mathbf{A}_{CD}\mathbf{R}^{-1}$$

$$\mathbf{A}_{RD} = \begin{bmatrix} 1 & 1 & 0 \\ j & -j & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -0.397 + j0.346 & 0 & 0 \\ 0 & -0.397 - j0.346 & 0 \\ 0 & 0 & -7.206 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ j & -j & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1}$$

$$\mathbf{A}_{RD} = \begin{bmatrix} -0.397 & 0.346 & 0 \\ -0.346 & -0.397 & 0 \\ 0 & 0 & -7.206 \end{bmatrix}$$

$$\mathbf{B}_{RD} = \mathbf{R}\mathbf{B}_{CD} = \begin{bmatrix} 1 & 1 & 0 \\ j & -j & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

$$\mathbf{C}_{RD} = \mathbf{C}_{CD}\mathbf{R}^{-1} = \begin{bmatrix} 0.153 - j1.367 & 0.153 + j1.367 & 4.694 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ j & -j & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1}$$

$$\mathbf{C}_{RD} = \begin{bmatrix} 0.153 & -1.367 & 4.694 \end{bmatrix}$$

$$\mathbf{D}_{RD} = \mathbf{D}_{CD} = 0$$

Use MATLAB to confirm that all the step responses are identical. Note that there might be a slight difference due to the rounding of numbers.

2. $\dfrac{Y(s)}{U(s)} = \dfrac{5s^2+7s+9}{s^2+2s+15}$

Follow the same steps shown in Q1.

***n*th order:**

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ -15 & -2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}-3\\ -60\end{bmatrix}u$$

$$y = \begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + 5u$$

CCF:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ -15 & -2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ 1\end{bmatrix}u$$

$$y = \begin{bmatrix}-66 & -3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + 5u$$

OCF:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & -15\\ 1 & -2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}-66\\ -3\end{bmatrix}u$$

$$y = \begin{bmatrix}0 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + 5u$$

DCF:

Again, this case is not required since the eigenvalues are -1±*j*3.742.

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}-1+j3.742 & 0\\ 0 & -1-j3.742\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}1\\ 1\end{bmatrix}u$$

$$y = \begin{bmatrix}-1.5+j8.419 & -1.5-j8.419\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + 5u$$

The real diagonal representation can be given as:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix}=\begin{bmatrix}-1 & 3.742\\ -3.742 & -1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}+\begin{bmatrix}2\\ 0\end{bmatrix}u$$

$$y=\begin{bmatrix}-1.5 & 8.419\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}+5u$$

3. $\dfrac{Y(s)}{U(s)}=\dfrac{3s+1}{s^3+6s^2+11s+6}$

Follow the same steps shown in Q1.

***n*th order:**

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix}=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -6 & -11 & -6\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}+\begin{bmatrix}0\\ 3\\ -17\end{bmatrix}u$$

$$y=\begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

CCF:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix}=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -6 & -11 & -6\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}+\begin{bmatrix}0\\ 0\\ 1\end{bmatrix}u$$

$$y=\begin{bmatrix}1 & 3 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

OCF:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix}=\begin{bmatrix}0 & 0 & -6\\ 1 & 0 & -11\\ 0 & 1 & -6\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}+\begin{bmatrix}1\\ 3\\ 0\end{bmatrix}u$$

$$y = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

DCF:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} -4 & 5 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

4. $\dfrac{Y(s)}{U(s)} = \dfrac{s+1}{s^2+5s+6}$

Follow the same steps shown in Q1.

***n*th order:**

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ -4 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

CCF:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

OCF:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & -6 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

DCF:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -3 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

5. $\dfrac{Y(s)}{U(s)} = \dfrac{4s+2}{s^3+4s^2+5s+2}$

Follow the same steps shown in Q1.

***n*th order:**

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -5 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 4 \\ -14 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

CCF:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -5 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 2 & 4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

OCF:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix}=\begin{bmatrix}0 & 0 & -2\\ 1 & 0 & -5\\ 0 & 1 & -4\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}+\begin{bmatrix}2\\ 4\\ 0\end{bmatrix}u$$

$$y=\begin{bmatrix}0 & 0 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

JCF:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix}=\begin{bmatrix}-p_1 & 1 & 0\\ 0 & -p_1 & 0\\ 0 & 0 & -p_3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}+\begin{bmatrix}0\\ 1\\ 1\end{bmatrix}u$$

$$y=\begin{bmatrix}c_1 & c_2 & c_3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}+b_0u$$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix}=\begin{bmatrix}-1 & 1 & 0\\ 0 & -1 & 0\\ 0 & 0 & -2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}+\begin{bmatrix}0\\ 1\\ 1\end{bmatrix}u$$

$$y=\begin{bmatrix}-2 & 6 & -6\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$